

1 Mission Objective and Scoring Function

The objective of the mission is to **maximize scientific return** from a long-duration tour of the **Mythara exoplanetary system** using propellant-less gravity assists. Scientific return is quantified through the following objective function, which must be maximized:

$$J = b \sum_{k \in ID} w_k \sum_{i=1}^{N_k} [S(\hat{\mathbf{r}}_{k,i}) \times F(V_{\infty,k,i})]$$

where:

- **b** is the *grand tour bonus term* (see [Section 3.1](#))
- **k** is the body index specified in [Table 1](#)
- **i** refers to the *i-th scientific flyby* (in chronological order) of body **k**
- **N_k** is the total number of scientific flybys of body **k**, with $N_k \leq 13$, i.e., up to 13 flybys per body may be designated as scientific and contribute to the score. Each flyby in the submitted solution is accompanied by a flag specifying whether it is scientific and should be counted toward the objective function (see the separate solution format file). Additional non-scientific flybys are permitted for navigation purposes but do **not** contribute to the score.
- **w_k** is the constant scientific weight of body **k**, reflecting its perceived scientific value
- **r_{k,i}** is the **unit heliocentric position vector** of body **k** at its *i*-th scientific flyby
V_{∞,k,i} is the **hyperbolic excess velocity magnitude** of the spacecraft relative to body **k** at the *i*-th scientific flyby
- **S** is the *seasonal penalty term* ([Section 3.2](#)), encouraging seasonal diversity.
- **F** is the *flyby velocity penalty term* ([Section 3.3](#)), rewarding slow, high-quality encounters

Example

If a tour is submitted at the start of the competition with a **single scientific flyby of Xerion** (the outer eccentric giant) and a relative flyby velocity of 10 km/s, then:

- **b=1** (no grand tour bonus)
- **w₁₀=50, N₁₀=1**
- **S=1**
- **F=0.663369**

Resulting in: **J=33,168**

Table 1 – Body Scientific Weights (Mythara System)

Body ID	Body Name	Weight w
4	Cindrel	3
1000	Nyxar	5
5	Aurelios	7
6	Tharos	10
7	Glacior	15
8	Ashkara	20
9	Varuun	35
10	Xerion	50

Note: Ambiguous nature of Nyxar (possibly a natural dwarf planet or an artificial alien megastructure) motivates its elevated scientific weight despite negligible mass.

2 Tie-Breaking Rules

If two solutions result in identical values of J to three decimal places:

1. The solution with the **greater total number of scientific flybys** prevails
 2. If still tied, the solution with the **larger initial incoming velocity** wins
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3 Parameters description

3.1 Grand Tour Bonus b

The grand tour bonus is defined as:

b=1.2 if the solution includes scientific flybys of all planets, and the dwarf planet Nyxar.
1.0 otherwise.

3.2 Seasonal Penalty Term S

To encourage **seasonal diversity** during repeated flybys of the same body, the seasonal penalty term is defined as:

$$S(\hat{\mathbf{r}}_{k,i}) = 0.1 + \frac{0.9}{1 + 10 \sum_{j=1}^{i-1} \exp\left(-\frac{[\text{acosd}(\hat{\mathbf{r}}_{k,i} \cdot \hat{\mathbf{r}}_{k,j})]^2}{50}\right)}$$

where:

- $\mathbf{r}_{k,i}$ is the unit heliocentric position vector of body k at its i -th scientific flyby
- **acosd** is the arccosine function with output expressed in degrees between 0° and 180° .
- $S(\mathbf{r}_{k,1}) = 1$

This term reduces the contribution of the i th scientific flyby of body k if its heliocentric direction $\hat{\mathbf{r}}_{k,i}$ is too similar to those of previous scientific flybys of that same body. Flybys of the same body clustered near the same solar phase angle are penalized, while those distributed across a broad range of viewing geometries are unaffected. This reflects the scientific value of observing a body under varying illumination conditions around Mythara.

3.3 Flyby Velocity Penalty Term F

The flyby velocity penalty term is defined as:

$$F(V_\infty) = 0.2 + \frac{\exp(-V_\infty/13)}{1 + \exp[-5(V_\infty - 1.5)]}$$

where V_∞ is expressed in km/s.

This function:

- Penalizes **high-velocity flybys**, which reduce observation time
- Penalizes **very low-velocity, rendezvous-like encounters**, due to radiation risk and environmental uncertainty near each body.

Coordinate Frame and Initial State of the Spacecraft

The ecliptic plane is the orbital plane of the accretion disk of the star. The initial heliocentric velocity of the spacecraft, at a distance of 200 AU from **Mythara**, is parallel to the ecliptic plane. Thus, the X direction is taken as lying along this velocity direction, positive towards

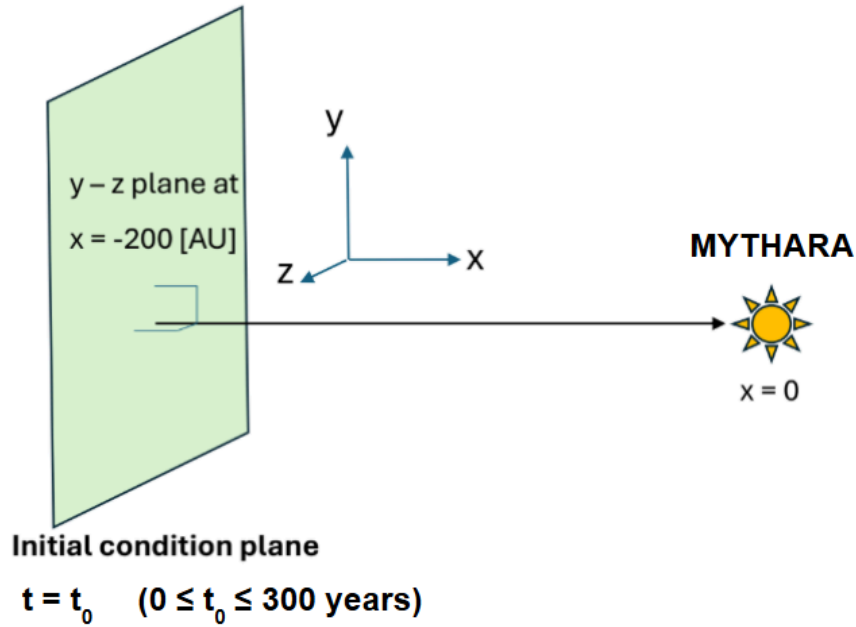
Mythara. The Z direction is taken as perpendicular to the ecliptic, positive in the direction of **the accretion disk** orbital angular momentum. The Y-axis is then defined as $\hat{y} = \hat{z} \times \hat{x}$. The orbital elements listed in the ephemeris files, as well as the initial spacecraft state listed in [Table 2](#), are all expressed in this coordinated frame. The solution files must also use this coordinate frame. For reference, the conversion between orbital elements and cartesian elements is given in Appendix 1.

The initial state of the spacecraft is defined in [Table 2](#) and illustrated in the figure below. It corresponds to an incoming interstellar asymptote nearly aligned with the +X direction. Initial position components are free in the YZ plane perpendicular to the X axis. This initial state is defined at initial time $0 \leq t_0 \leq 300$ years, i.e. the initial time is not fixed but must be selected between 0 and 300 years. It is recommended (but not mandatory) to start with a positive V_x value.

The Mean Anomaly listed in the ephemeris files is for reference epoch time $t=0$.

Table 2 – Initial spacecraft state

Initial States	Value
X	-200 AU
Y	Free
Z	Free
V_x	Free
V_y	0
V_z	0



Constants

The values of the constant parameters of the problem are provided in [Table 4](#).

Table 4 – constants

Constant	Value
AU	149597870.691 km
Mythara GM (μ)	139348062043.343 km ³ /s ²
Day	86400 s
Year	365.25 days

7. Constraints, Tolerances and Solution Checking

1. All the trajectory events (including trajectory start & flybys) must occur within a fixed time window of 300 years measured from the reference epoch $t = 0$. In particular, the initial time t_0 when the initial conditions in [Table 2](#) are defined must satisfy $0 \leq t_0 \leq 300$ years. Similarly, the last time of the trajectory solution should lie between 0 and 300 years.
2. All close approaches to **Mythara** except for one must be at or above a range of 0.05 AU. The spacecraft is equipped with a thermal protection system which will allow a

single perihelion passage as low as 0.01 AU. This single, lower passage can be applied at any perihelion, i.e. it does not need to be the first perihelion passage.

3. If any two successive flybys (scientific or non-scientific) are of the same body, then the time interval between these flybys must be no less than 1/3 of the body orbital period around **Mythara**, in order to allow enough time for navigation of each flyby.
4. The heliocentric position vector of the spacecraft at the time of a flyby of a body must be equal to the heliocentric position vector of the body at that time (subject to conic position tolerance given below).
5. Planetary flybys (body ID between 4 and 10) are modeled using the gravity assist patched conics model. The incoming and outgoing spacecraft hyperbolic excess velocities relative to the planet must have equal magnitude. Each flyby must occur at an altitude between 0.1 and 100 body radii from the surface of the flyby body. The flyby altitude is computed from the patched conics model.
6. The dwarf planet **Nyxar** is treated as massless. Therefore, the incoming and outgoing spacecraft hyperbolic excess velocities relative to this body must have equal magnitude and direction (i.e. be continuous) at the flyby of this body.

Upon submission via the competition website, solutions will be checked against an independent propagations generated from the submitted data. Solutions will have to meet the following tolerances to be considered valid:

Table 5 - Tolerances

Parameter	Tolerance / Value
Conic Position & Velocity tolerance	100 m; 0.1 mm/s
Initial (trajectory start) Position & Velocity tolerance	100 m; 0.1 mm/s
V^∞ equality tolerance	0.1 mm/s
Time, Position & Velocity continuity tolerance	To all reported digits of accuracy
Relative tolerance for numerical integration (applies to both position and velocity over 1 propagated segment)	10^{-4}

Reporting interval for numerical integration	> 60 s (intervals much longer than 60 s are generally expected)
Flyby-altitude tolerance	100 m
Perihelion altitude tolerance	1 km